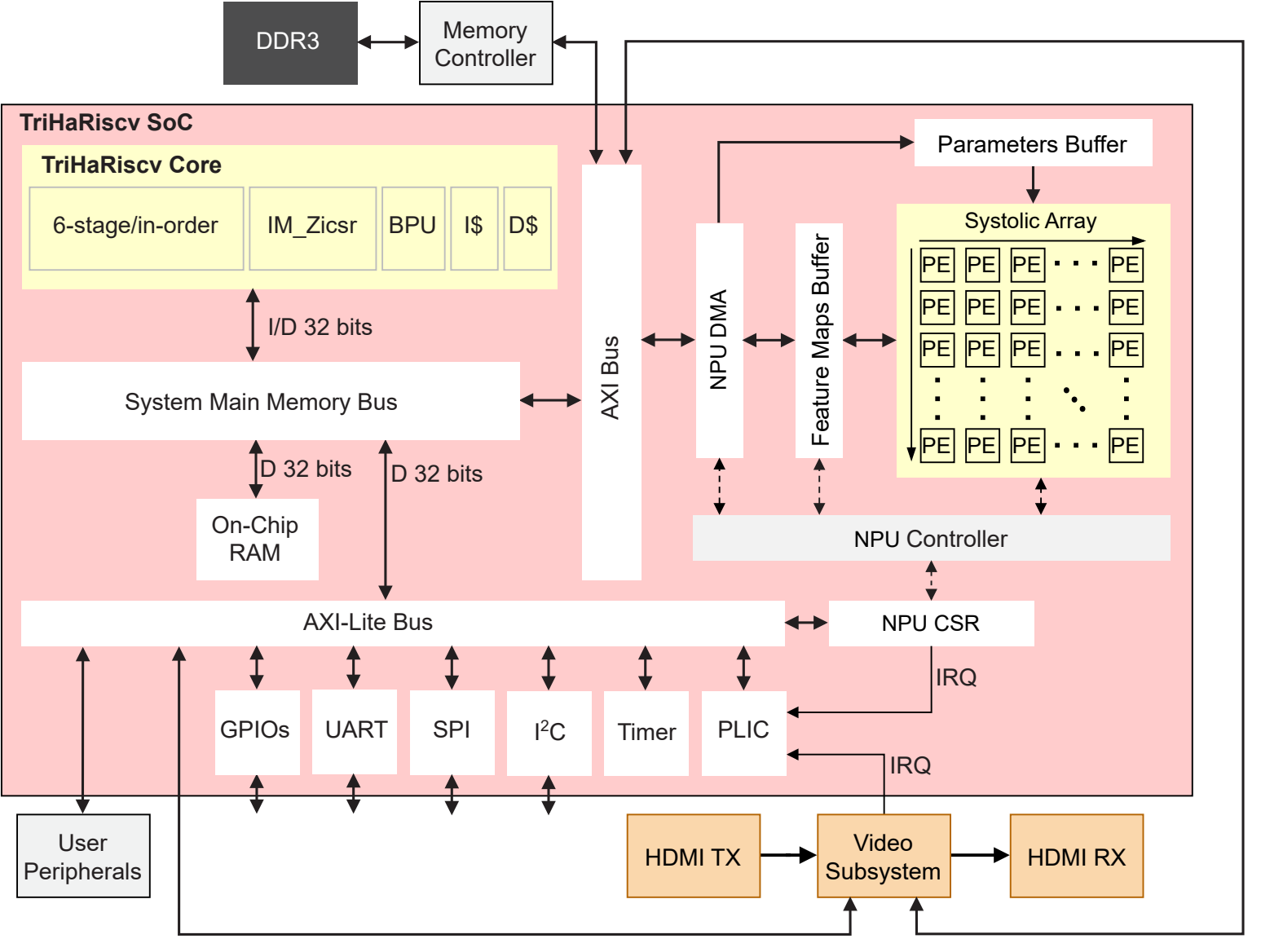
DDR3
Memory Controller
TriHaRiscv SoC
TriHaRiscv Core
6-stage/in-order
IM_Zicsr
BPU
I$
D$
I/D 32 bits
System Main Memory Bus
D 32 bits
D 32 bits
On-Chip RAM
AXI Bus
NPU DMA
Feature Maps Buffer
Parameters Buffer
Systolic Array
PE
NPU Controller
AXI-Lite Bus
NPU CSR
IRQ
GPIOs
UART
SPI
I2C
Timer
PLIC
IRQ
User Peripherals
HDMI TX
Video Subsystem
HDMI RX